PGCP-AI

PRACTICAL MACHINE LEARNING

Day 4: Regression and its types

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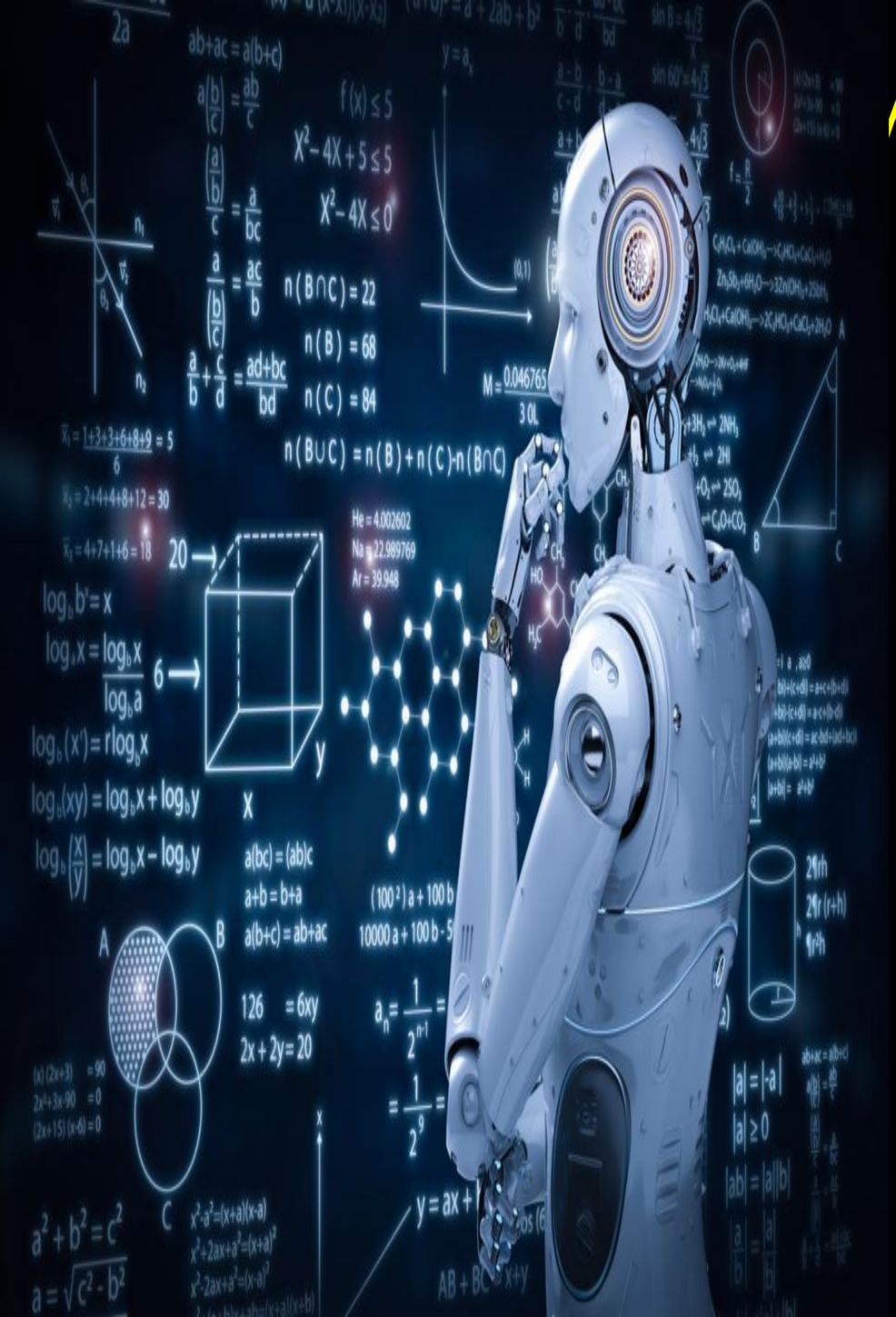
CDAC Mumbai



Agenda

- ML Algorithms
- Regression
- Types of Regression
- Ridge
- Lasso
- ElasticNet
- Logistic

Simple Linear Regression Equation (Prediction Line)

The simple linear regression equation provides an estimate of the population regression line

Estimated (or predicted) Y value for observation i

Estimate of the regression intercept

Estimate of the regression slope

Value of X for observation i

$$\hat{Y}_i = b_0 + b_1 X_i$$

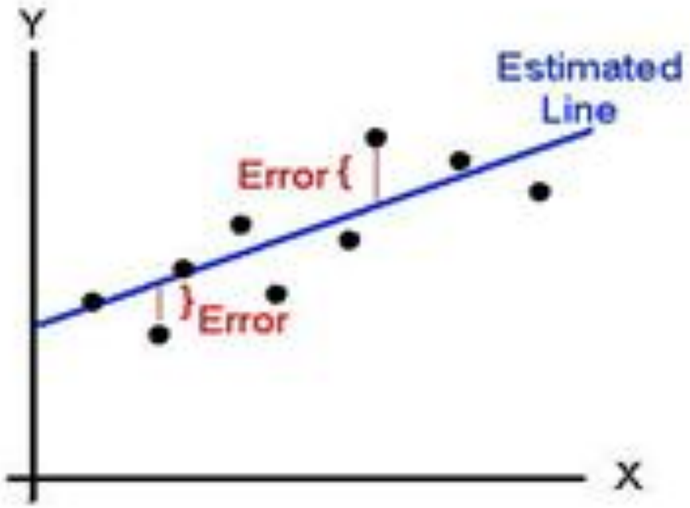
Estimated (or predicted) Y value for observation i

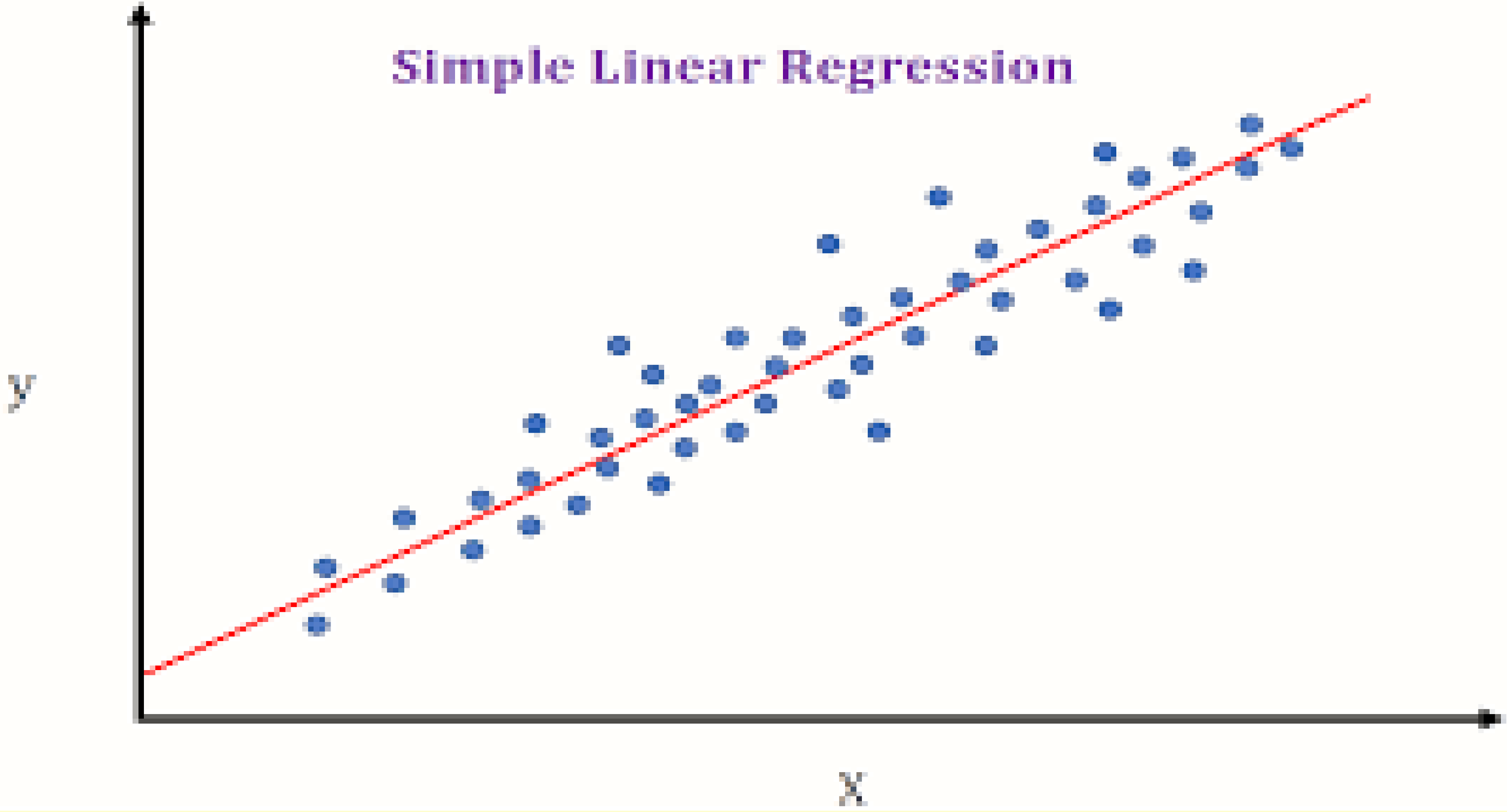
Estimate of the regression intercept

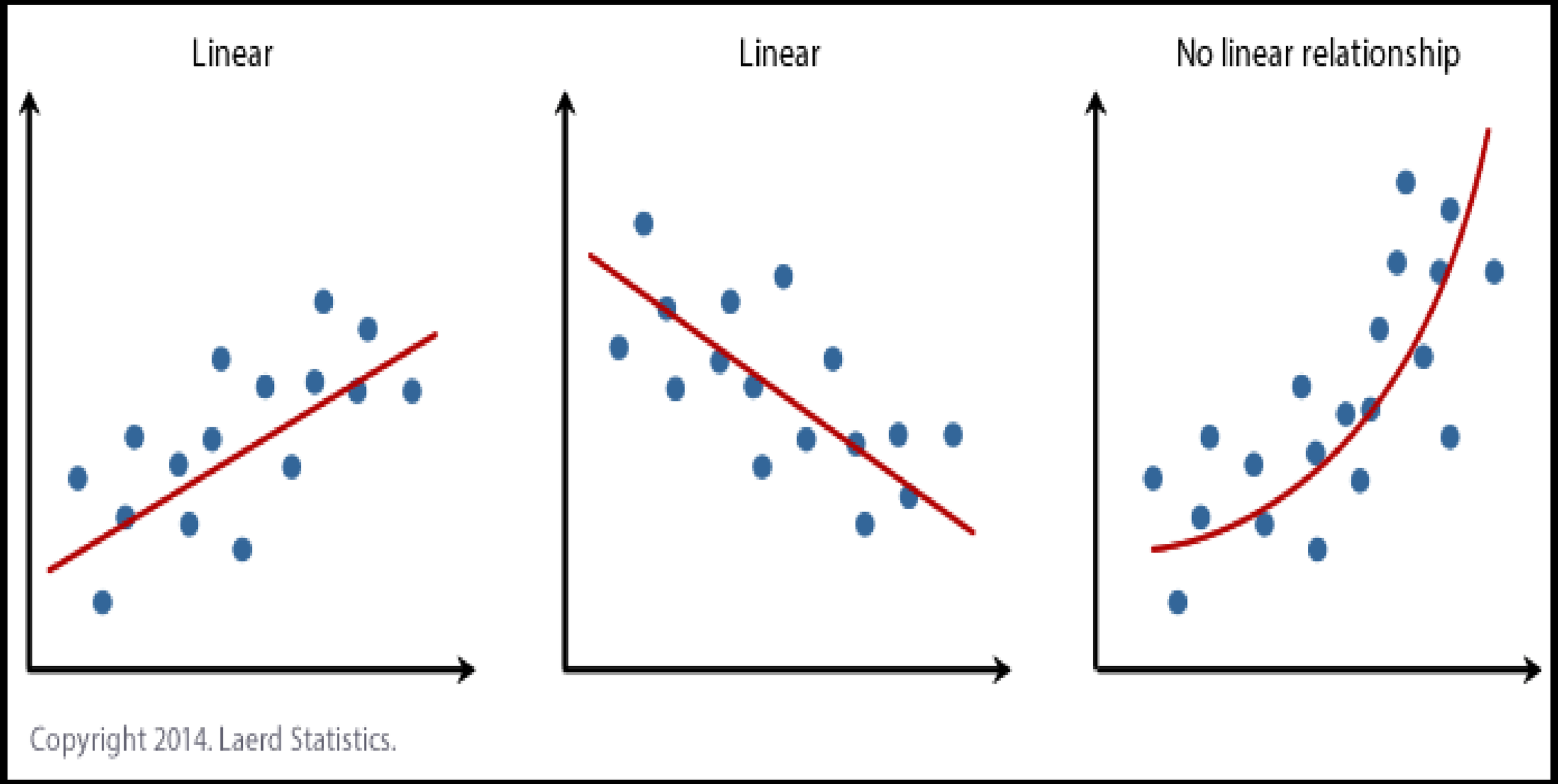
Estimate of the regression slope

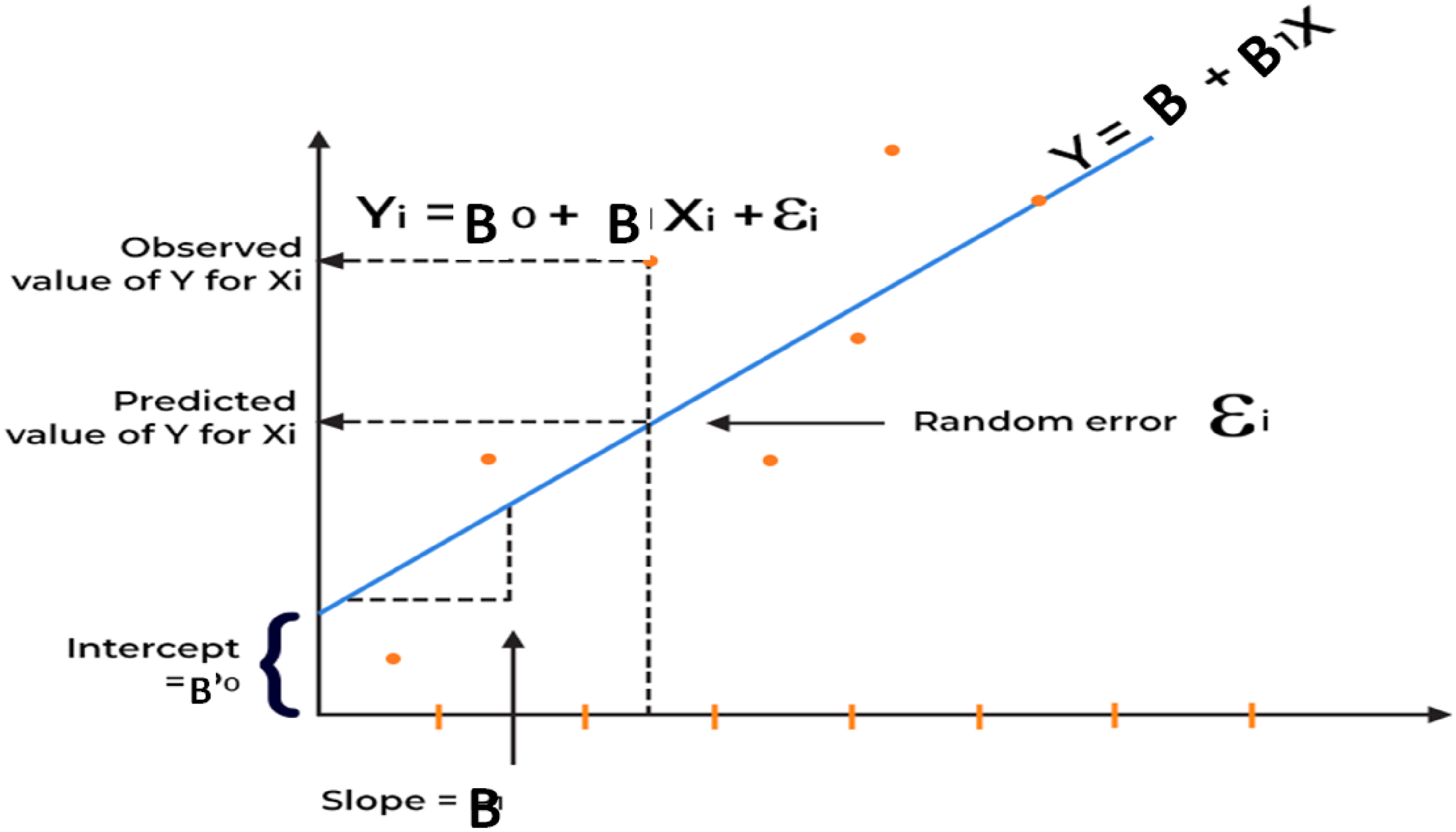
Value of X for observation i

$$\hat{Y}_i = b_0 + b_1 X_i$$




Simple Linear Regression
y
x


Linear
Linear
No linear relationship
Copyright 2014. Laerd Statistics.


Y = B + B1X
Yi = B o + B1 Xi + εi
Observed value of Y for Xi
Predicted value of Y for Xi
Random error εi
Intercept =B'o
Slope = B1

What is Multiple Linear Regressions?

Multiple Linear Regressions (MLR)

A statistical means used extensively to model relationship between variables

- It determines the relationship between one dependent and several independent variables
- Is an extension of simple linear regression
- It aids in making prediction and conclusion regarding the involved components
- The MLR model is based on ordinary least squares
- Helps making statistical prediction of components

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_i X_i$$

Y : Dependent variable
β_0 : Intercept
β_i : Slope for X_i
X = Independent variable

Dependent Variable
(Response Variable)

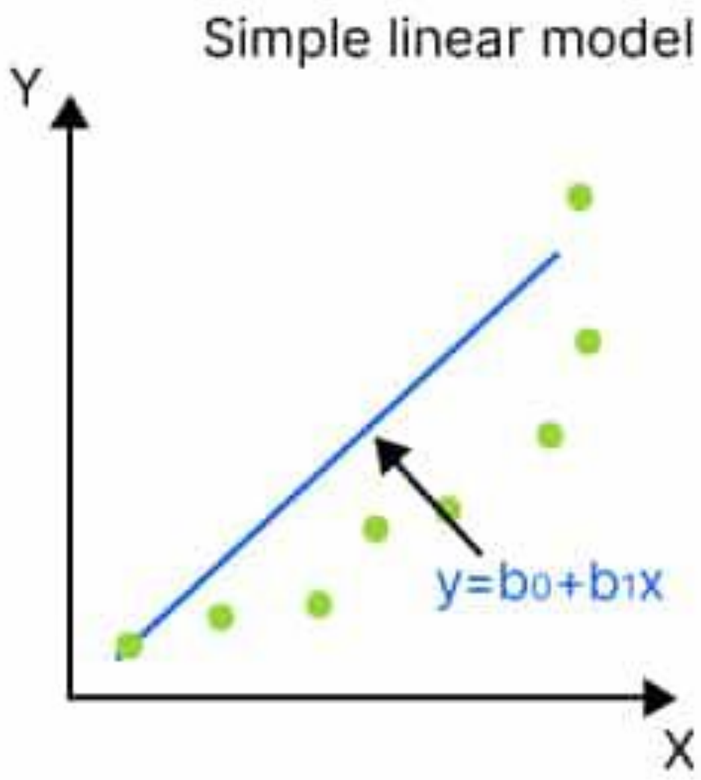
Independent Variables
(Predictors)

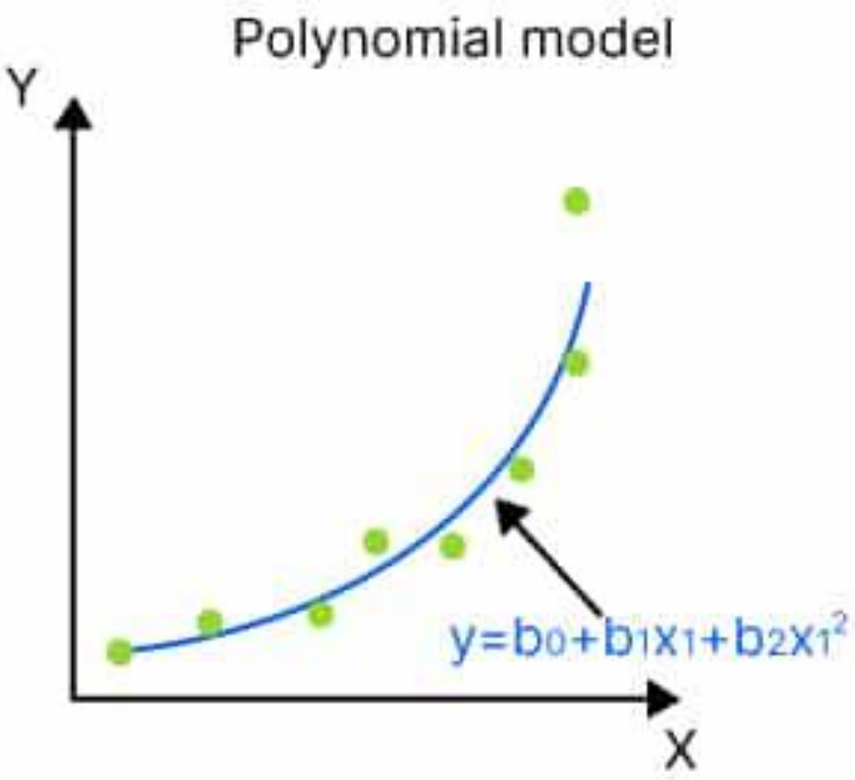
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \varepsilon$$

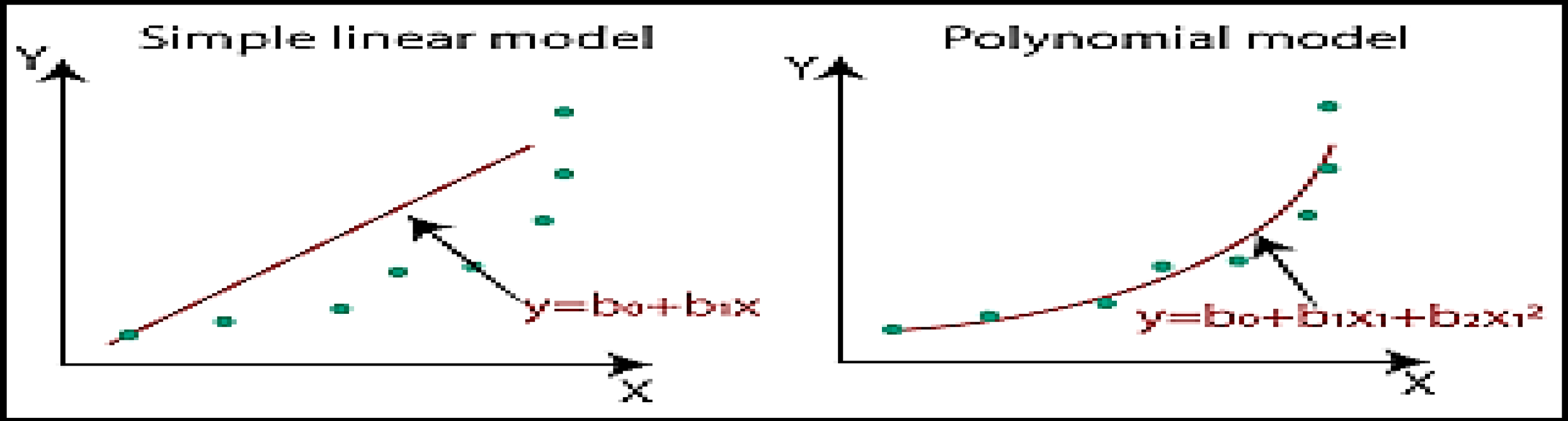
Y intercept

Slope
Coefficient

Error Term


Simple linear model
Y
X
y=b0+b1x


Polynomial model
Y
X
y=b0+b1x1+b2x1²



Simple Linear Regression

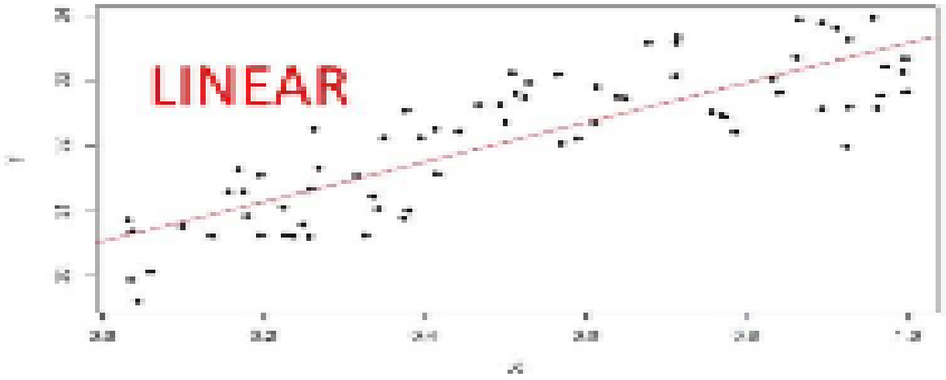
$$y = b_0 + b_1 x_1$$

Multiple Linear Regression

$$y = b_0 + b_1 x_1 + b_2 x_2 + \ldots + b_n x_n$$

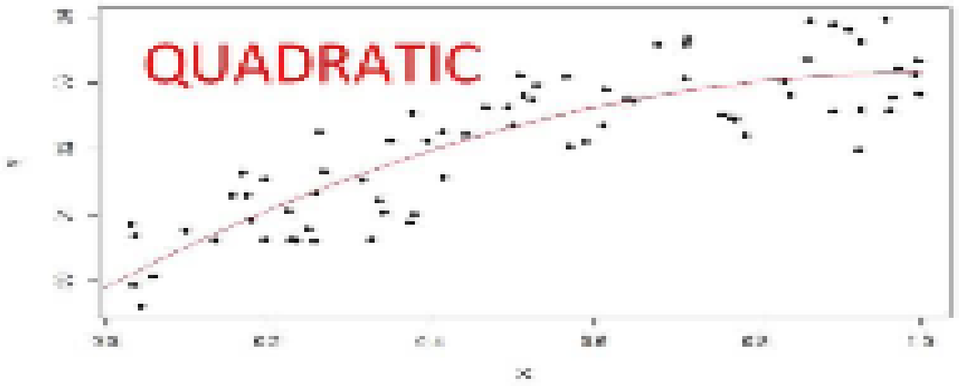
Polynomial Linear Regression

$$y = b_0 + b_1 x_1 + b_2 x_1^2 + \ldots + b_n x_1^n$$



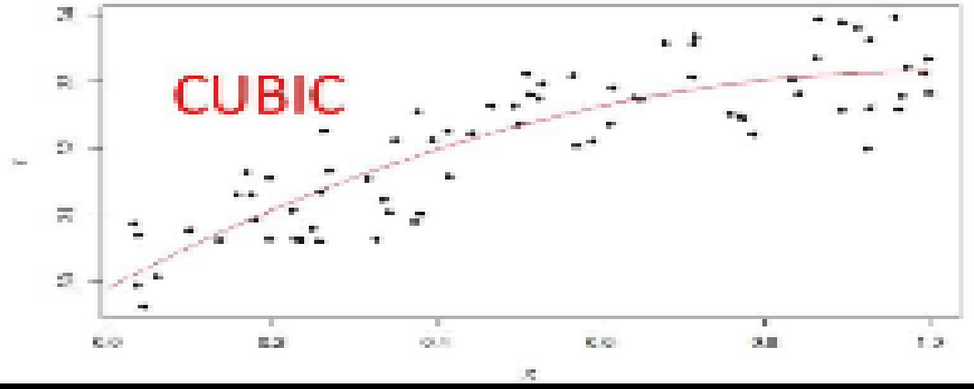
Multiple R-squared: 0.7044

$Y = 30.53 + 3.05*X$



Multiple R-squared: 0.7559

$Y = 29.90 + 6.48*X - 3.22*X^2$

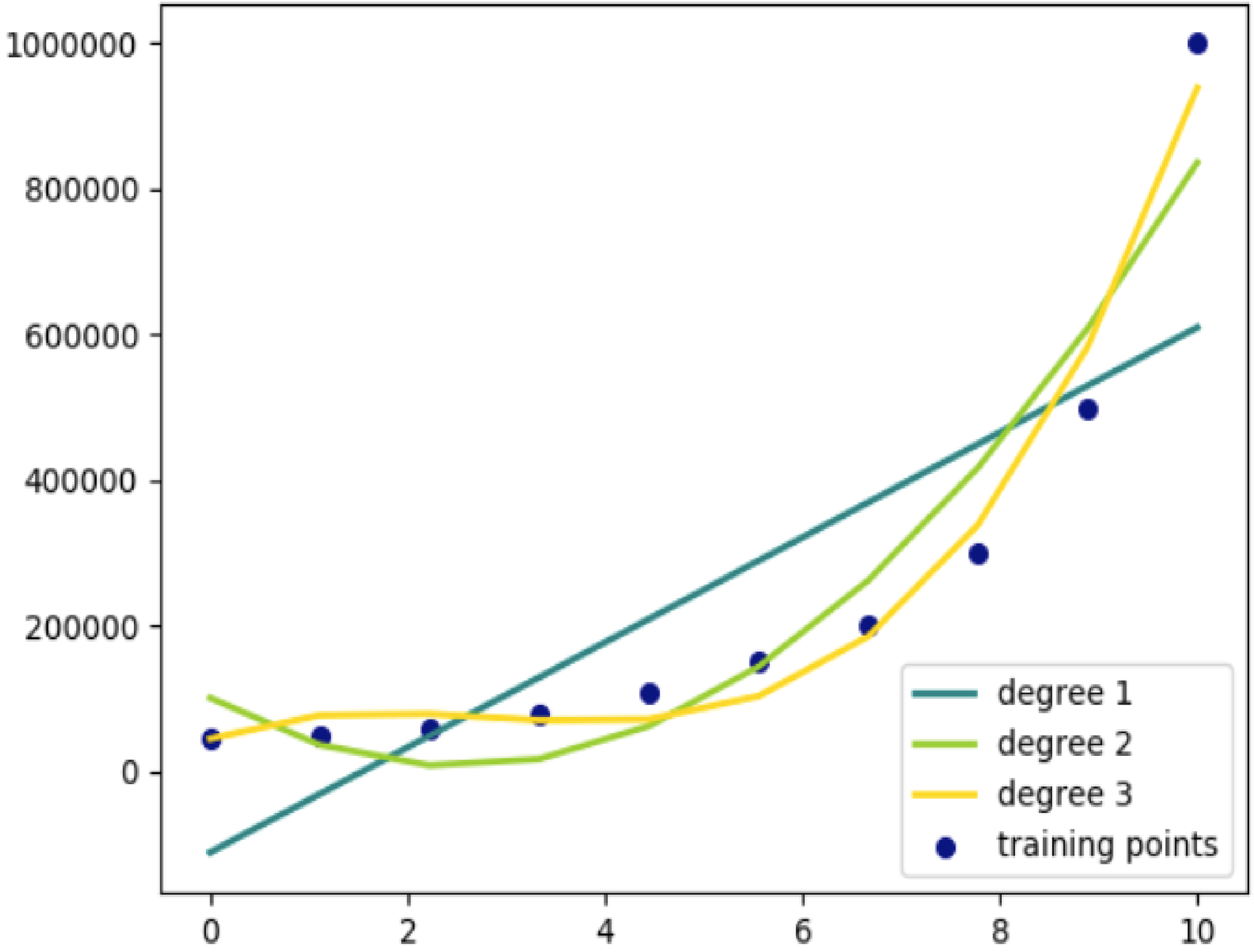


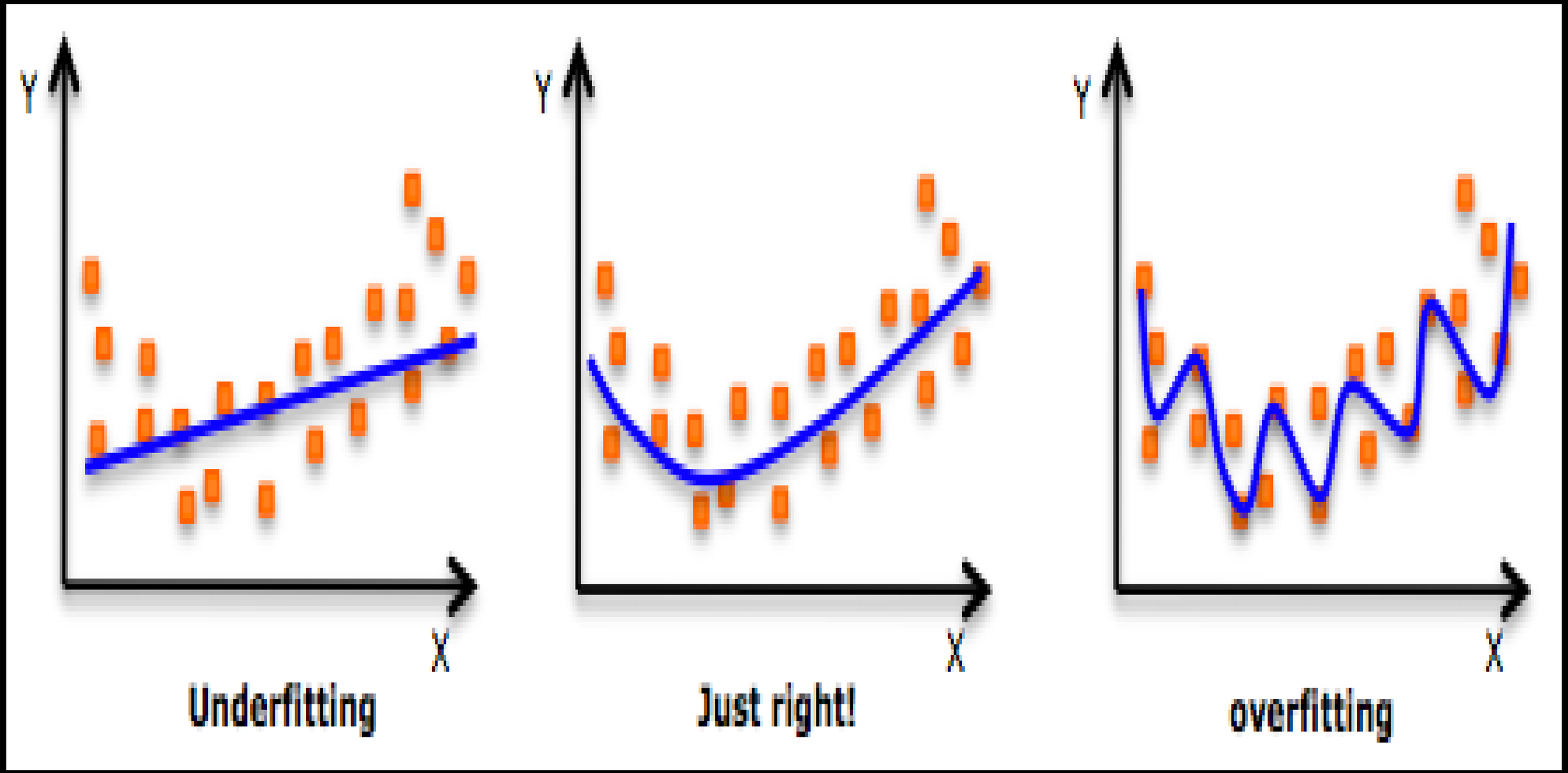
Multiple R-squared: 0.7623

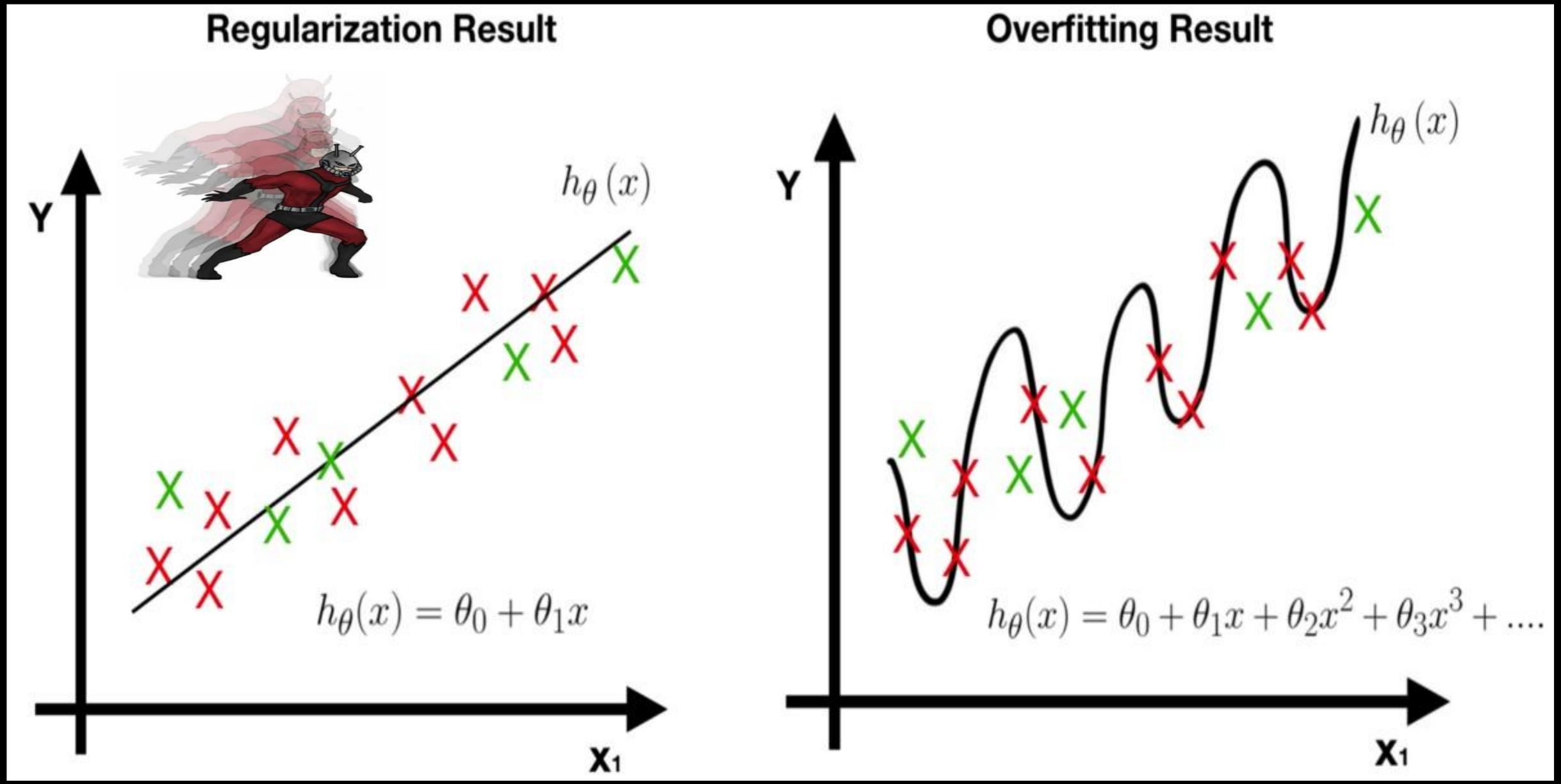
$Y = 30.17 + 3.61*X + 3.71*X^2 - 4.48*X^3$

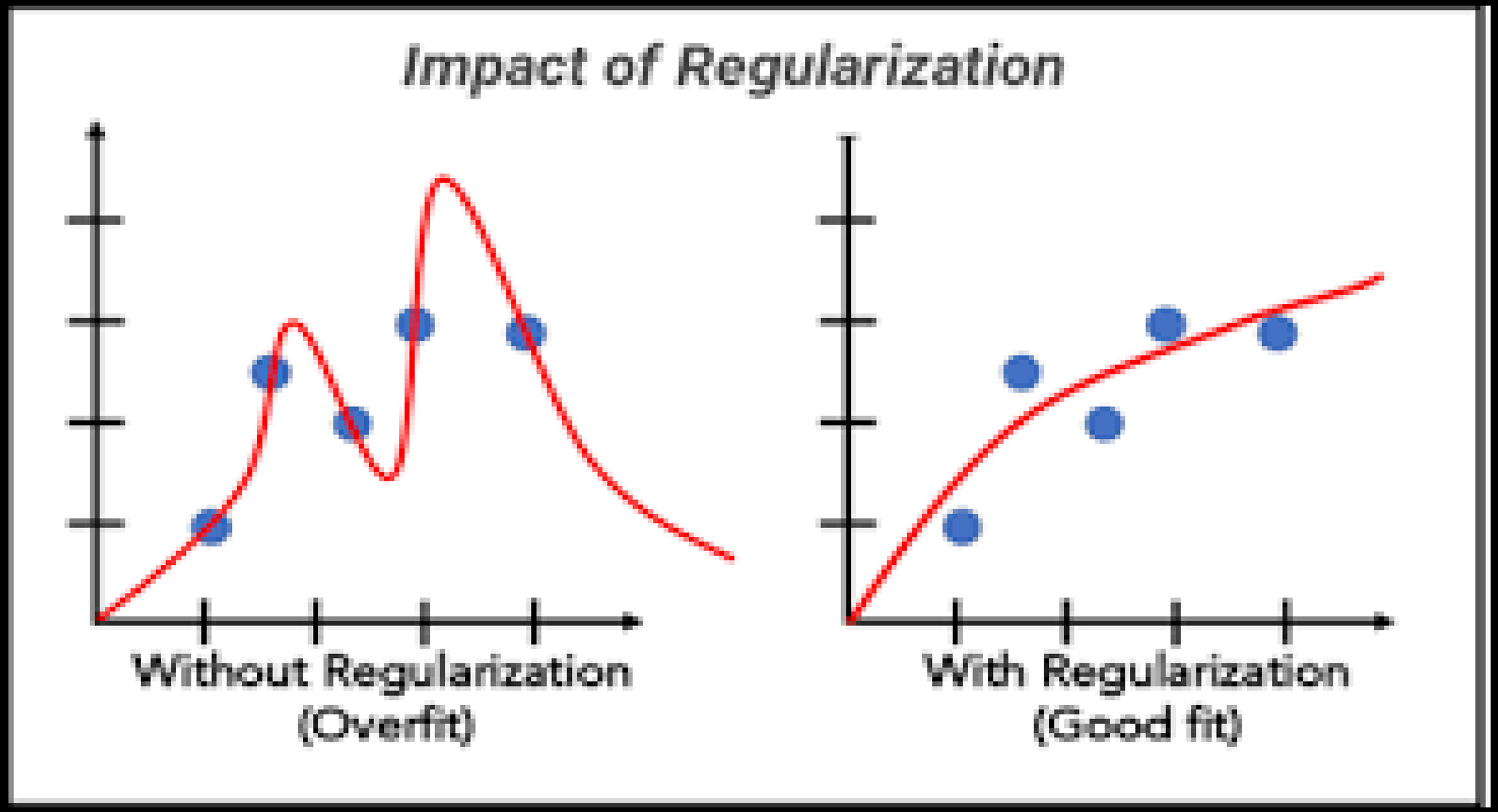
salarydata

Position	Level	Salary
Business Analyst	1	45000
Junior Consultant	2	50000
Senior Consultant	3	60000
Manager	4	80000
Country Manager	5	110000
Region Manager	6	150000
Partner	7	200000
Senior Partner	8	300000
C-level	9	500000
CEO	10	1000000




Y
X
Underfitting
Y
X
Just right!
Y
X
overfitting


Regularization Result
Y
$h_\theta(x)$
$h_\theta(x) = \theta_0 + \theta_1 x$
X_1
Overfitting Result
Y
$h_\theta(x)$
$h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + ...$
X_1


Impact of Regularization
Without Regularization
(Overfit)
With Regularization
(Good fit)

Regularization

- The loss function is the function that **computes the distance between the current output of the algorithm and the expected output** .
- It's a method to **evaluate how your algorithm models** the data.
- A loss function **specifies a penalty for an incorrect estimate from a statistical model**.

Regularization Term

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta(x^{(i)}) - y^{(i)}\right)^2 + \boxed{\boxed{\lambda}\sum_{j=1}^{n}\theta_j^{\,2}}$$

start at θ_1

Regularization Parameter

Lasso Regression

Lasso regression uses the same mean squared error loss function and this applies L1 Regularization and will repeat the same steps as Ridge. The cost function of Lasso Regression $J(\theta)$ is given as

$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m}(y - \hat{y})^2 + \lambda \sum_{j=1}^{n}|w_j|$$

where

$\lambda \sum_{j=1}^{n}|w_j|$ is the penalty (L1 Regularization).

LASSO, short for Least Absolute Shrinkage and Selection Operator, is a statistical formula whose main purpose is the feature selection and regularization of data models.

Where is lasso used?

Lasso regression is a regularization technique. It is used over regression methods for a more accurate prediction.

Ridge Regression

Ridge regression uses the mean squared error loss function and applies L2 Regularization. Its cost function *J(θ)* is given as

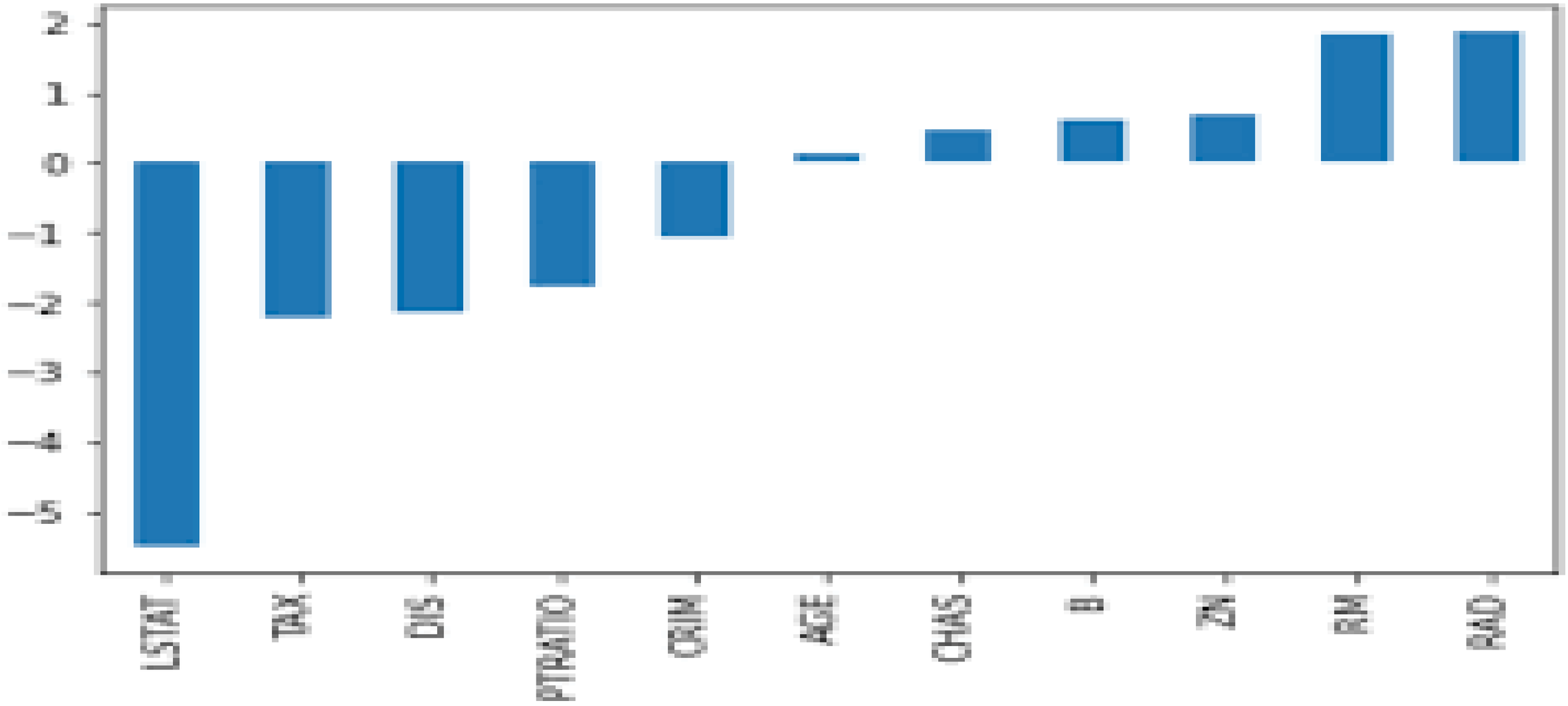
$$J(\theta) = \frac{1}{m}\sum_{i=1}^{m}(y - \hat{y})^2 + \lambda \sum_{j=1}^{n} w_j^2$$

where,

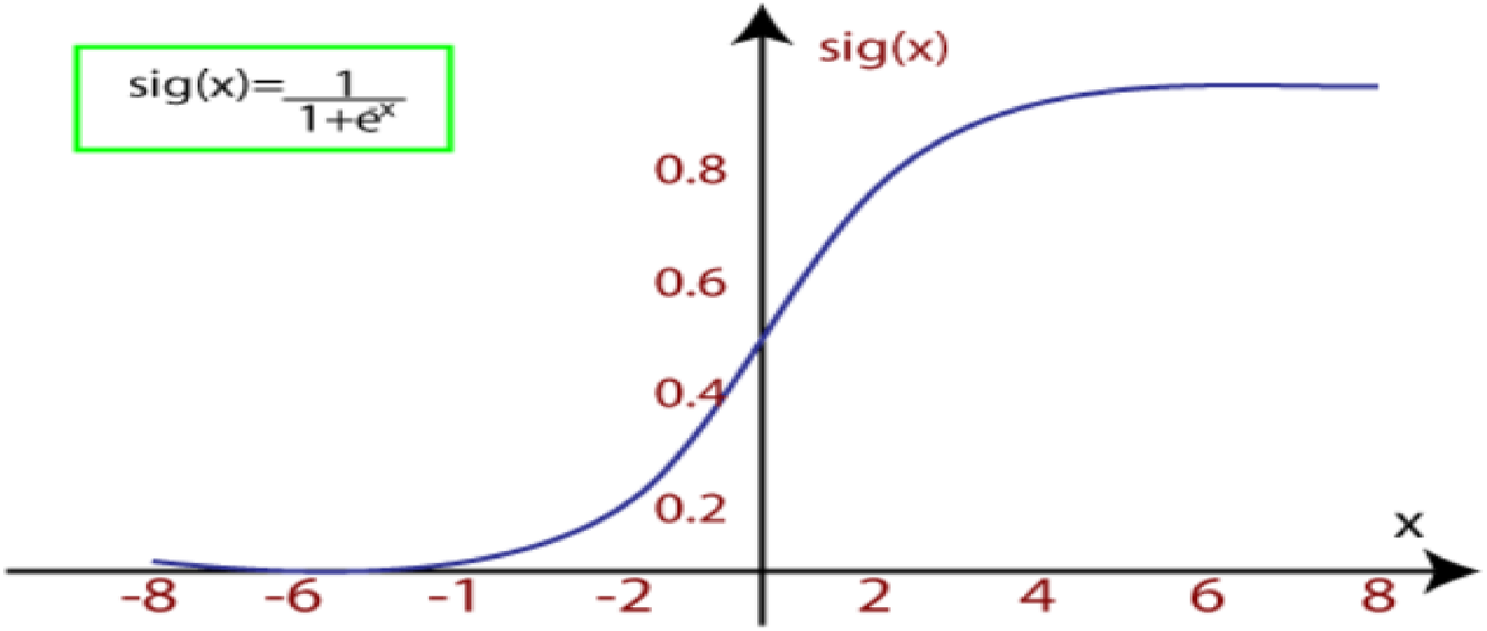
$\frac{1}{m}\sum_{i=1}^{m}(y - \hat{y})^2$ is the Mean Squared error (loss function)

$\lambda \sum_{j=1}^{n} w_j^2$ is the penalty (L2 Regularization)

Now, substitute $\hat{y}$ as $wx_i + b$.


2
1
0
−1
−2
−3
−4
−5
LSTAT
TAX
DIS
PTRATIO
CRIM
AGE
CHAS
B
ZN
RM
RAD

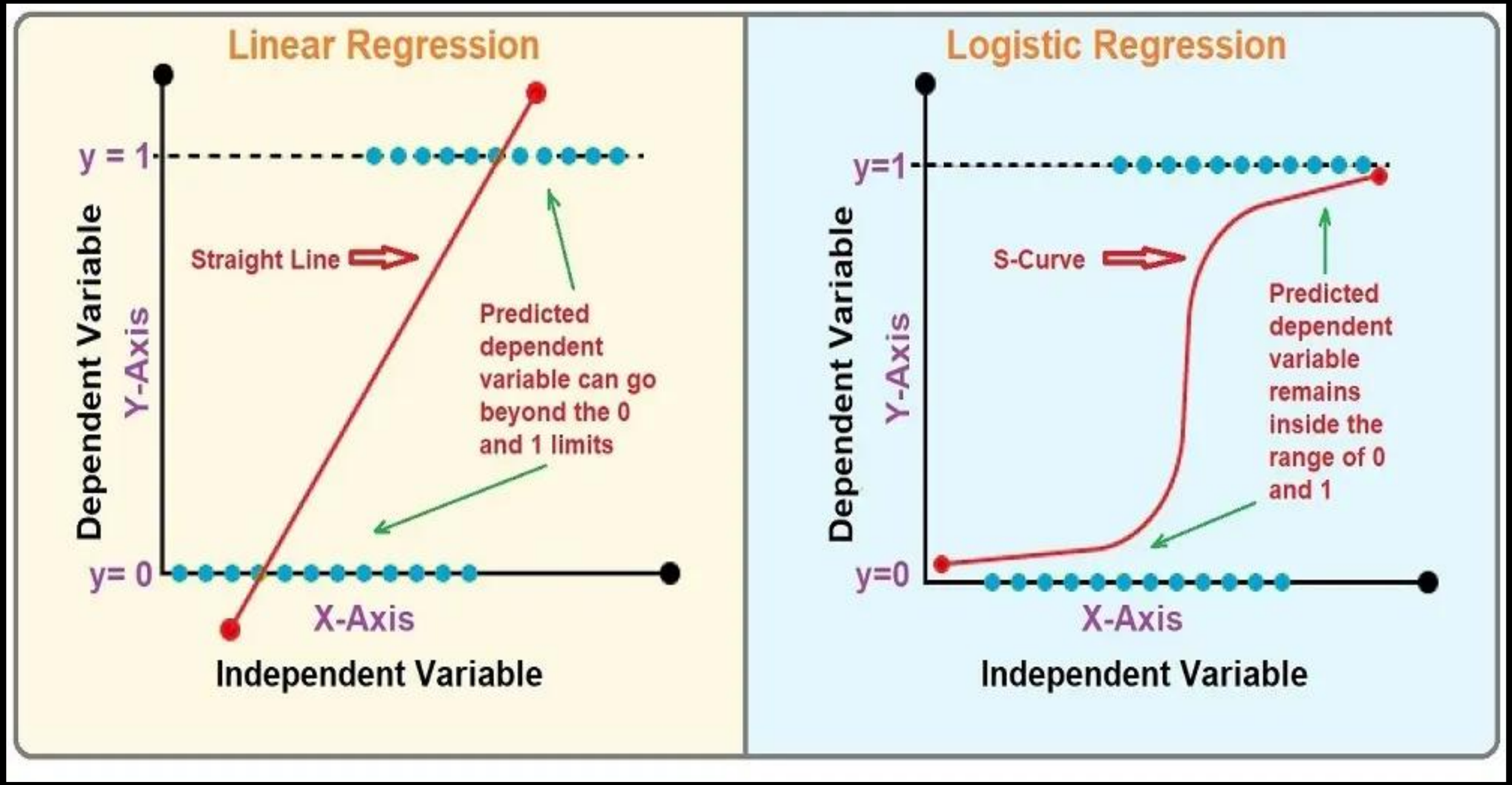
Logistic Regression



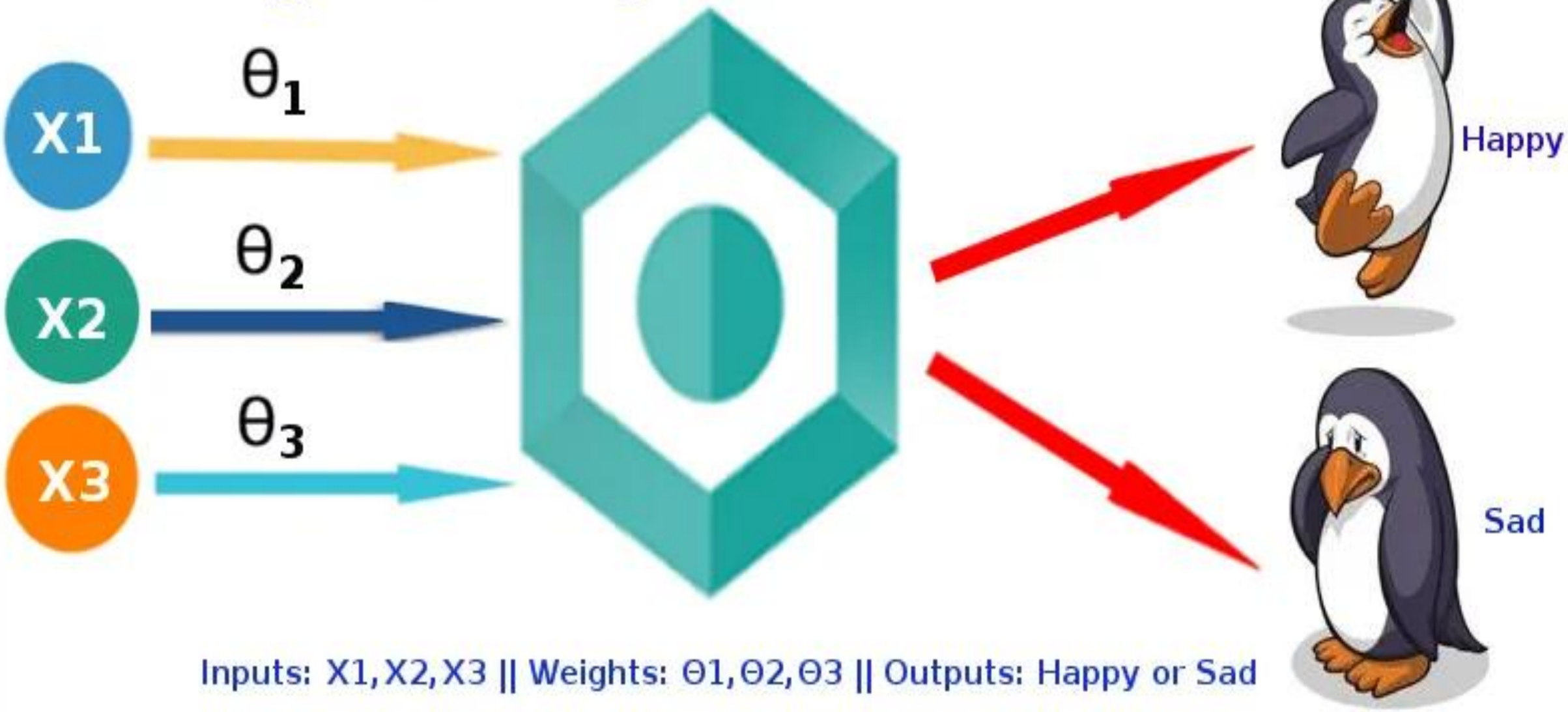
- It uses the concept of threshold levels, values above the threshold level are rounded up to 1, and values below the threshold level are rounded up to 0.

There are three types of logistic regression:

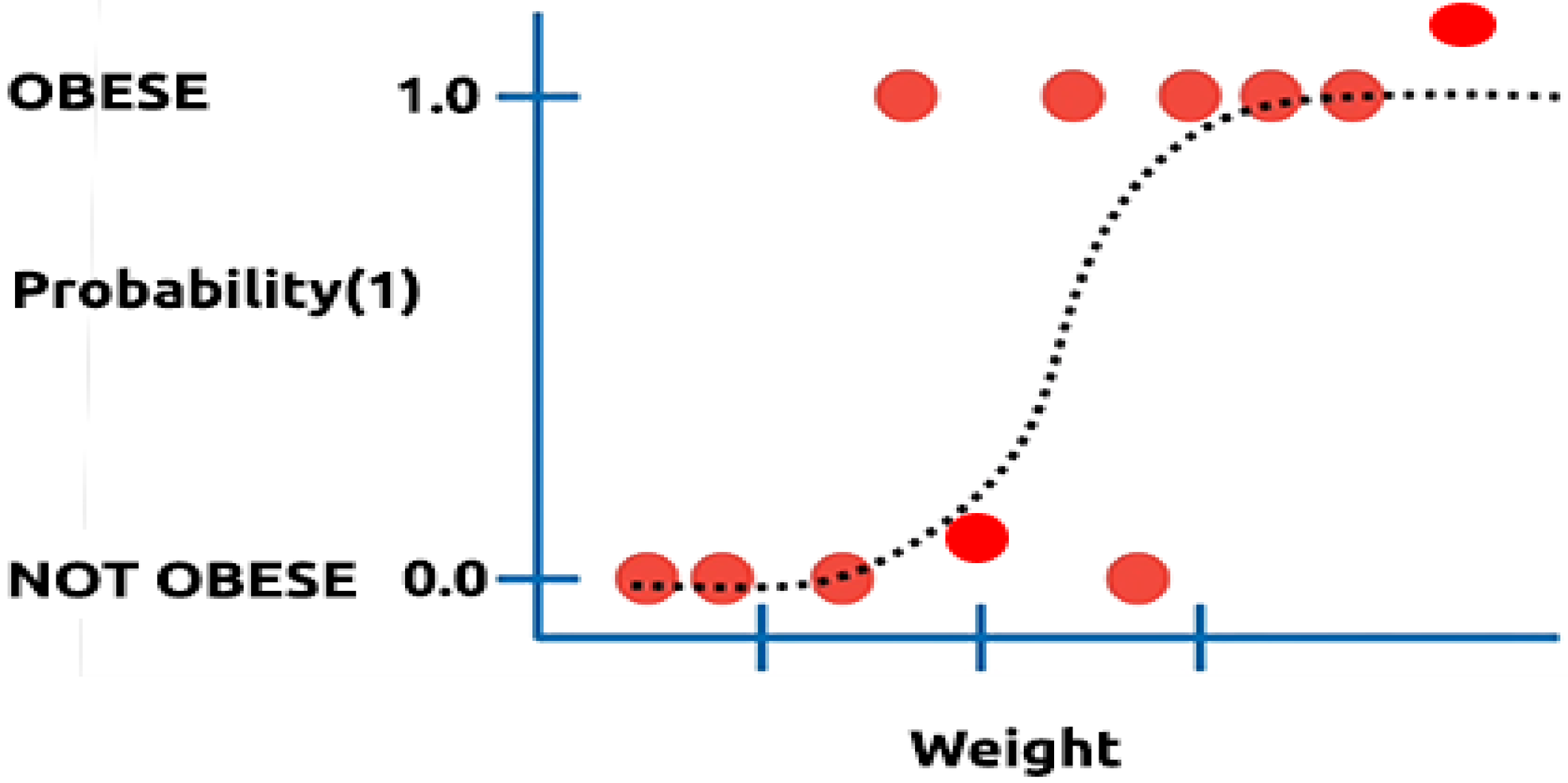
- **Binary(0/1, pass/fail)**
- **Multi(cats, dogs, lions)**
- **Ordinal(low, medium, high)**

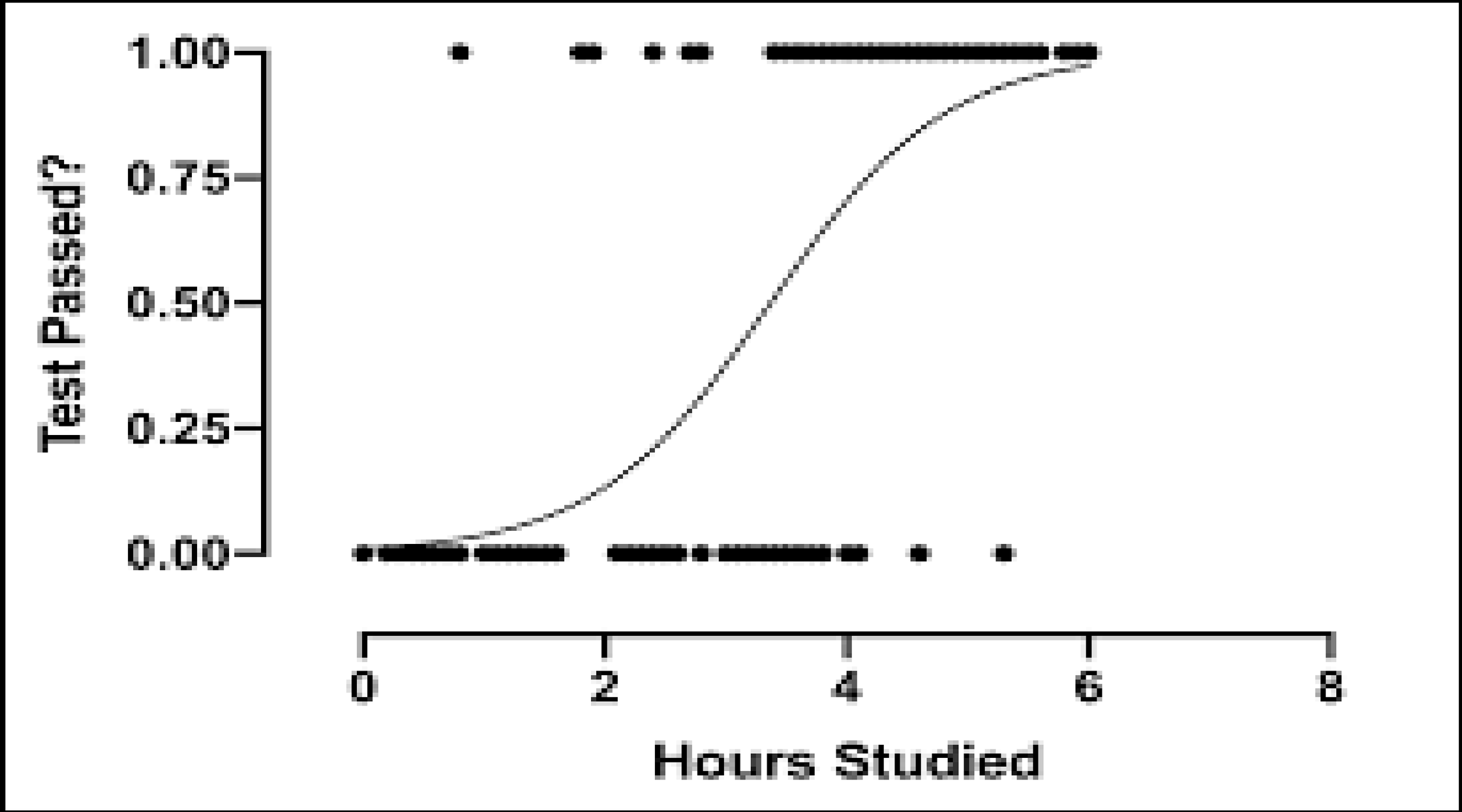

Linear Regression
y = 1
y= 0
Dependent Variable
Y-Axis
Straight Line
Predicted dependent variable can go beyond the 0 and 1 limits
X-Axis
Independent Variable
Logistic Regression
y=1
y=0
Dependent Variable
Y-Axis
S-Curve
Predicted dependent variable remains inside the range of 0 and 1
X-Axis
Independent Variable

Logistic Regression Model

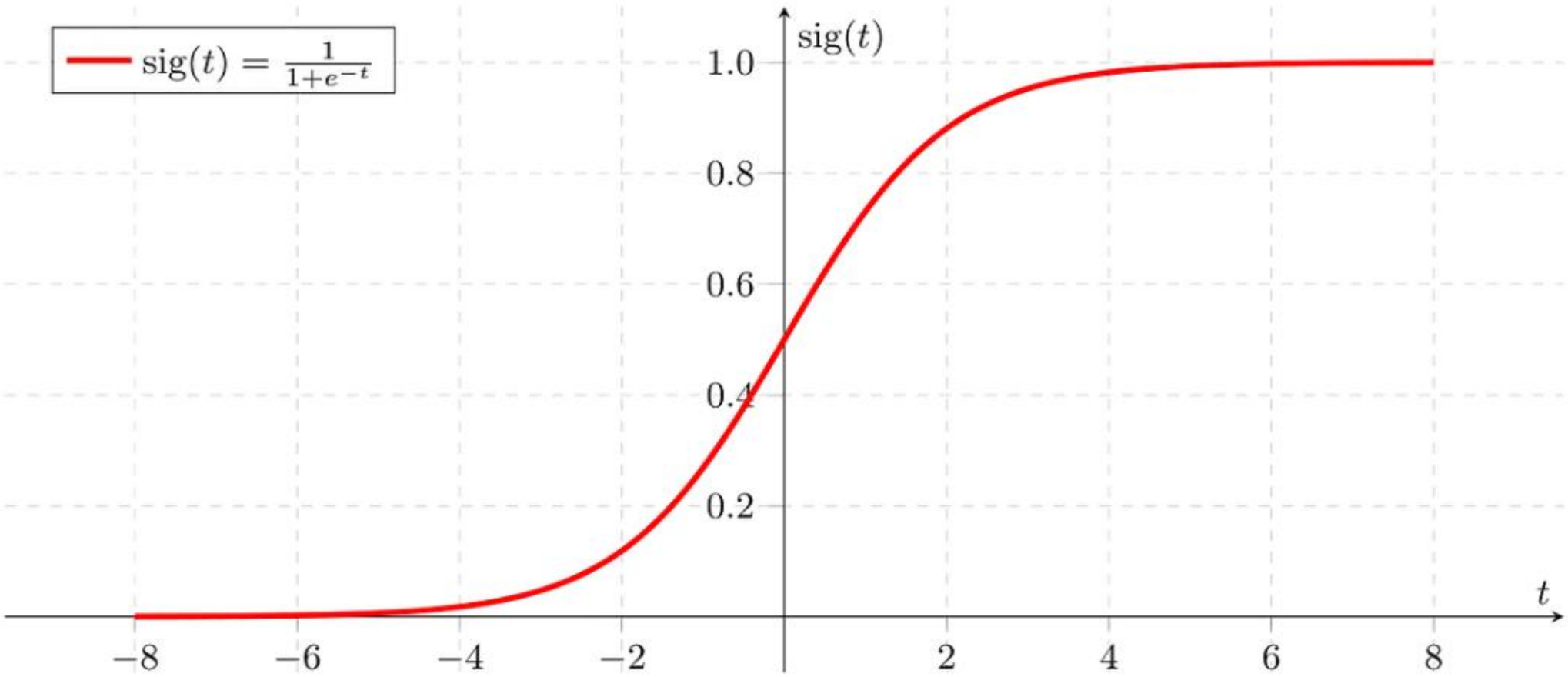


Inputs: X1, X2, X3 || Weights: Θ1, Θ2, Θ3 || Outputs: Happy or Sad

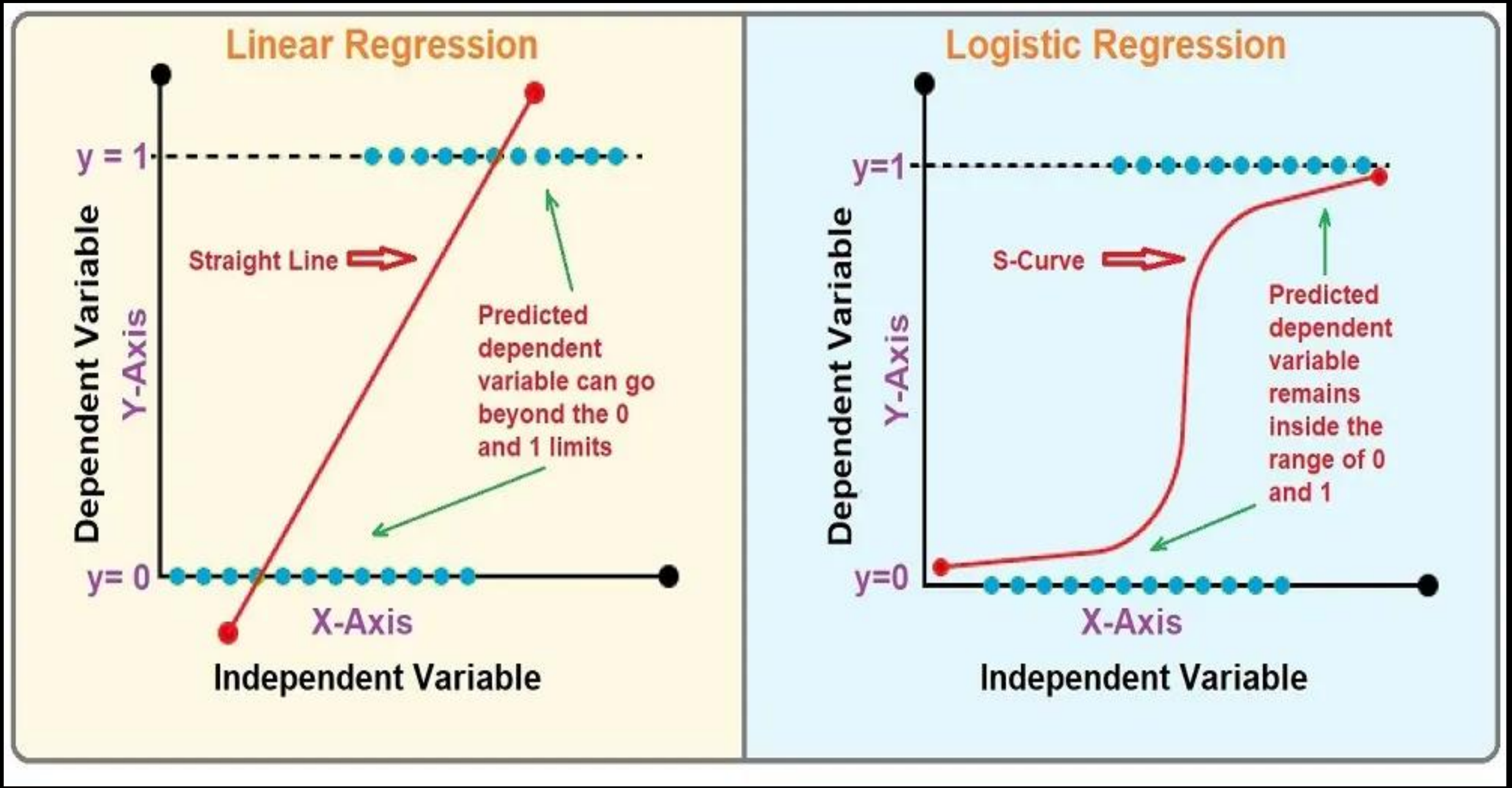

OBESE
1.0
Probability(1)
NOT OBESE
0.0
Weight


1.00
0.75
0.50
0.25
0.00
Test Passed?
0
2
4
6
8
Hours Studied

Sigmoid Function



Sigmoid Activation Function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Straight Line
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X-Axis
Independent Variable
Logistic Regression
y=1
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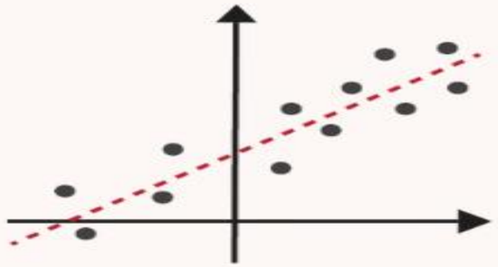
S-Curve
Predicted dependent variable remains inside the range of 0 and 1
X-Axis
Independent Variable

Linear regression VS Logistic regression

Linear regression

- Econometric modeling
- Marketing mix model
- Customer lifetime value



Continuous > Continuous

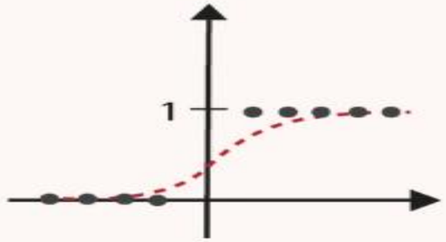
$$y = a_0 + \sum_{i=1}^{N} a_i x_i$$

Im(y-x1 + x2, data)

1 unit increase in
x increases y by a

Logistic regression

- Customer choice model
- Click-through rate
- Conversion rate
- Credit scoring



Continuous > True/False

$$y = \frac{1}{1 + e^{-z}}$$

$$z = a_0 + \sum_{i=1}^{N} a_i x_i$$

glm(y-x1 + x2, data), family = binomial())

1 unit increase in x
increases log odds by a

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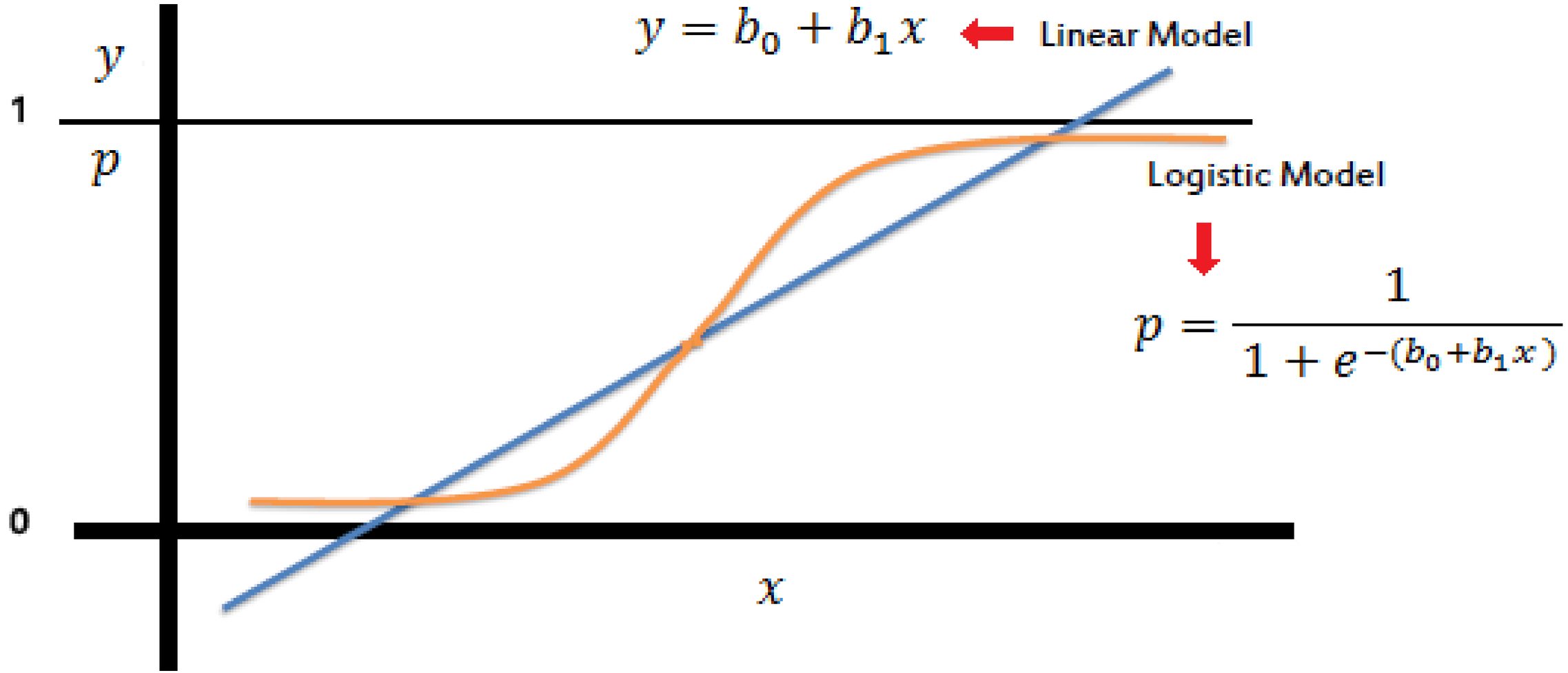

$y = b_0 + b_1 x$
Linear Model
y
1
p
Logistic Model
$p = \frac{1}{1 + e^{-(b_0 + b_1 x)}}$
0
x

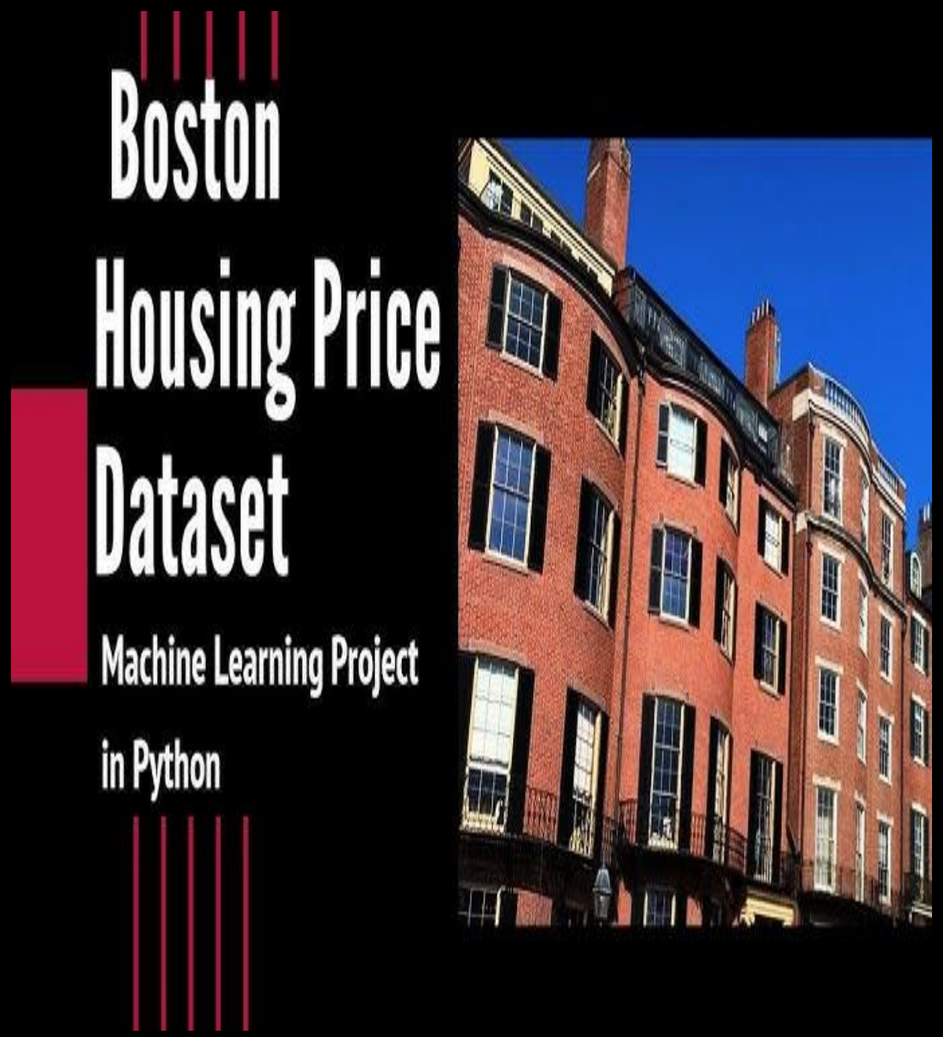
Iris dataset

- Many exploratory data techniques are nicely illustrated with the iris dataset.
  - Dataset created by famous statistician Ronald Fisher
  - 150 samples of three species in genus *Iris* (50 each)
    - *Iris setosa*
    - *Iris versicolor*
    - *Iris virginica*
  - Four attributes
    - sepal width
    - sepal length
    - petal width
    - petal length
  - Species is class label

Iris virginica. Robert H. Mohlenbrock. USDA NRCS. 1995. Northeast wetland flora: Field office guide to plant species. Northeast National Technical Center, Chester, PA. Courtesy of USDA NRCS Wetland Science Institute.

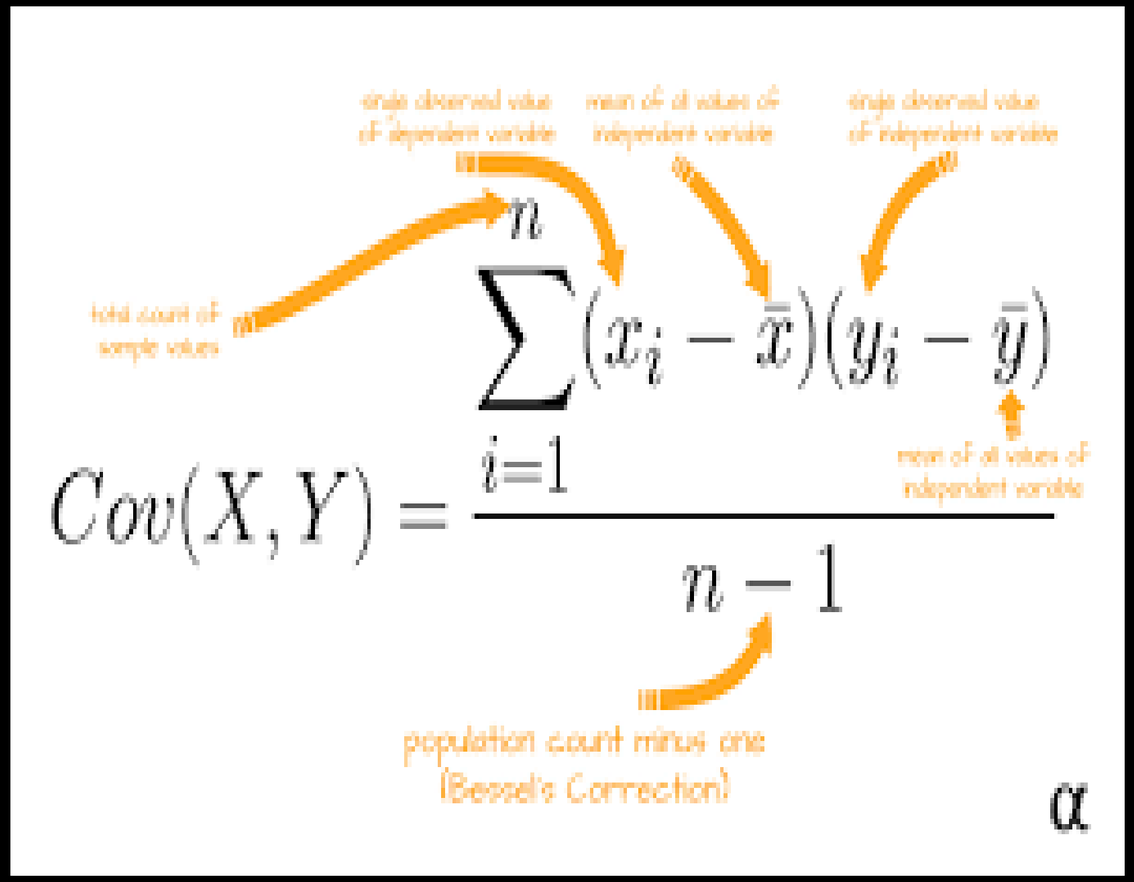
Boston Housing Dataset

- Boston Housing dataset has 506 samples with 13 features.
- It predicts house prices (medv).
- Used for regression modeling and ML practice.
- **Key Characteristics**
- Dataset contains both numerical and one binary feature (chas)
- Used primarily for:
  - Linear Regression
  - Feature selection
  - Model evaluation
- Some variables show strong correlation with house price
  - Positive: rm
  - Negative: lstat, ptratio



Covariance

Covariance indicates whether an increase in one variable corresponds to an increase or decrease in another variable.



Why covariance matters in ML ?

Feature relationship analysis

- Helps identify whether features move together

Feature selection

- Highly correlated/covariant features may be redundant

Dimensionality reduction (PCA)

- Covariance matrix is used to find principal components

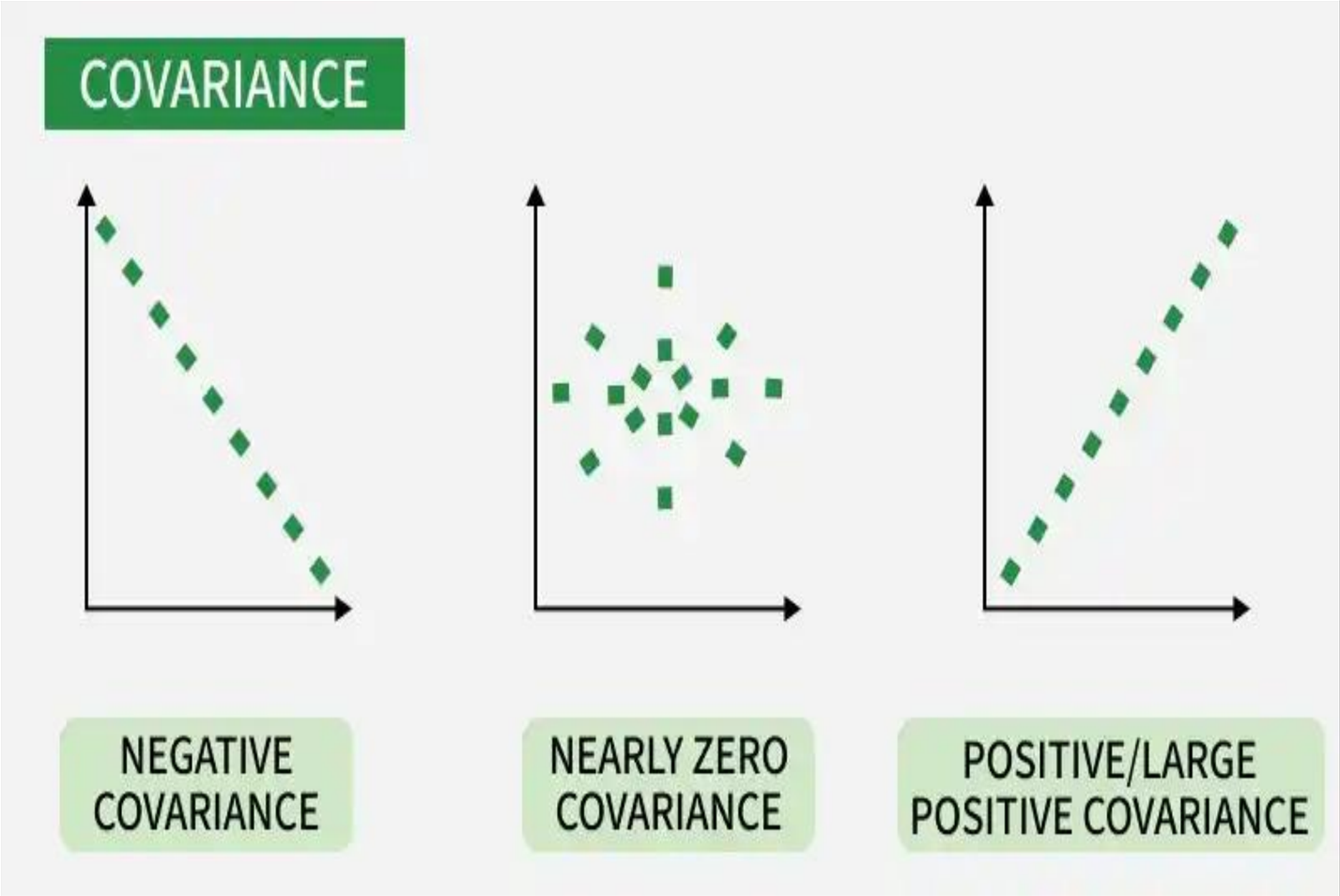
Data understanding

- Helps detect dependencies in datasets

Covariance

Interpretation

1. Positive covariance
   - If X increases, Y also tends to increase
   - Example: height and weight
2. Negative covariance
   - If X increases, Y tends to decrease
   - Example: speed and travel time
3. Zero covariance
   - No linear relationship between variables
   - Example: shoe size and IQ



Correlation

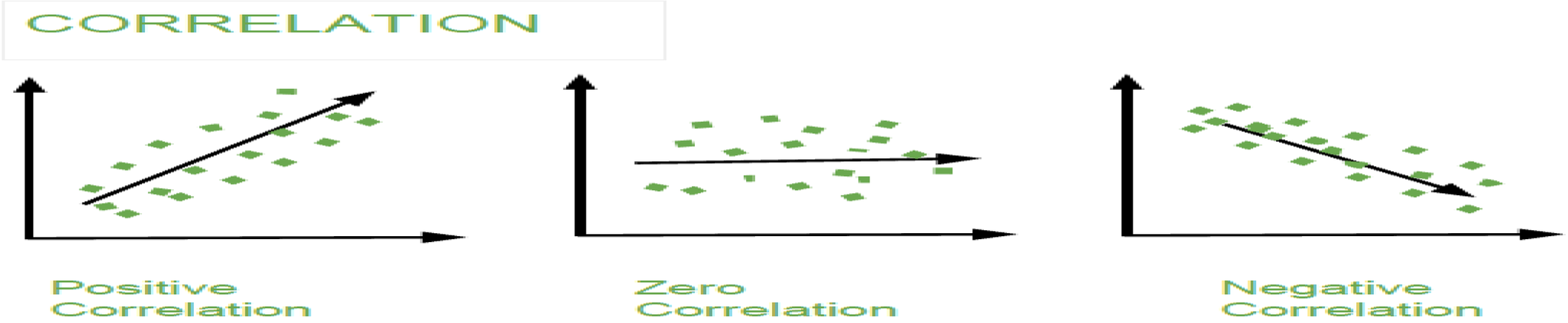
The most common form is Pearson correlation coefficient:

$$\rho_{xy} = \frac{Cov(x.y)}{\sigma_x \sigma_y}$$

Range and interpretation

Correlation always lies between **-1 and +1**:

Value	Meaning
+1	Perfect positive correlation (move together)
0	No linear relationship
-1	Perfect negative correlation (opposite movement)



Why correlation is important in ML

1. Feature selection

- Highly correlated features may be redundant
- Helps reduce multicollinearity in models

2. Data understanding

- Identifies relationships between variables

3. Model stability

- Reducing correlated inputs improves linear model performance

4. Dimensionality reduction

- Used in methods like PCA (indirectly through covariance matrix)

Types of correlation

1. Positive correlation

- One variable increases → other increases
- Example: height and weight

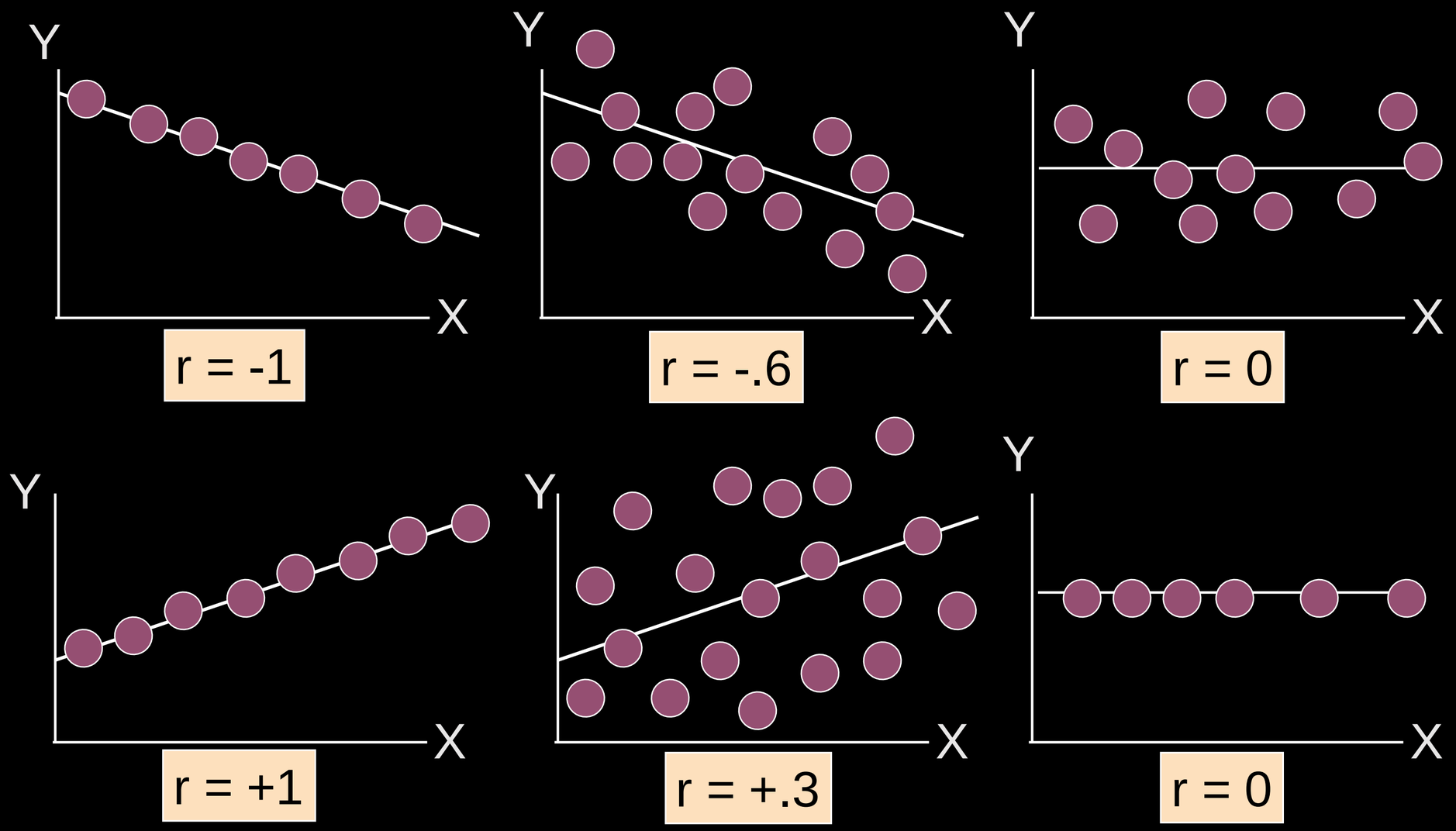
2. Negative correlation

- One increases → other decreases
- Example: speed and travel time

3. Zero correlation

- No linear relationship
- Example: shoe size and salary

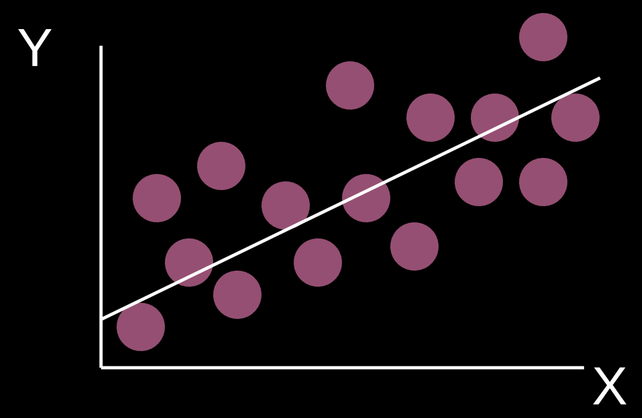
Scatter Plots of Data with Various Correlation Coefficients

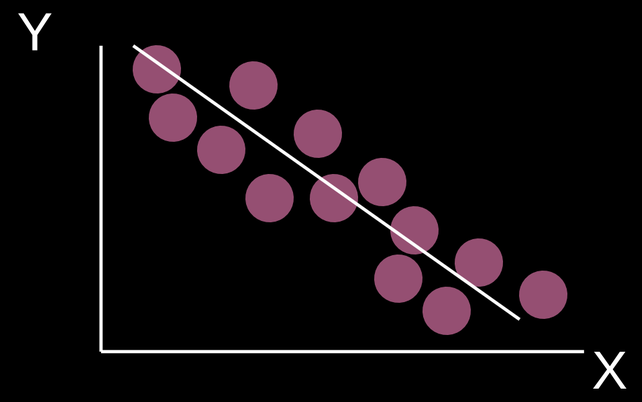


- Slide from: Statistics for Managers Using Microsoft® Excel 4th Edition, 2004 Prentice-Hall

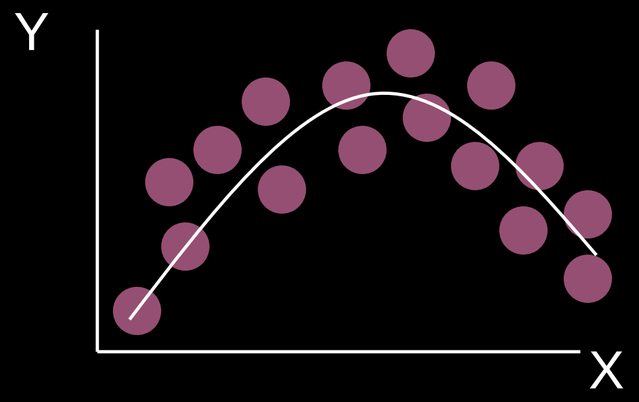
Linear Correlation

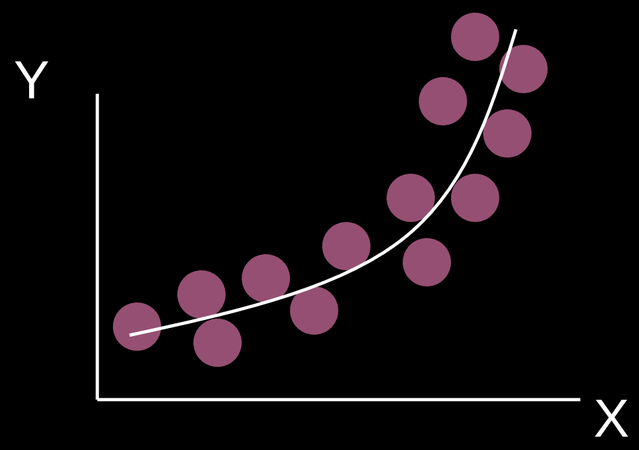
Linear relationships





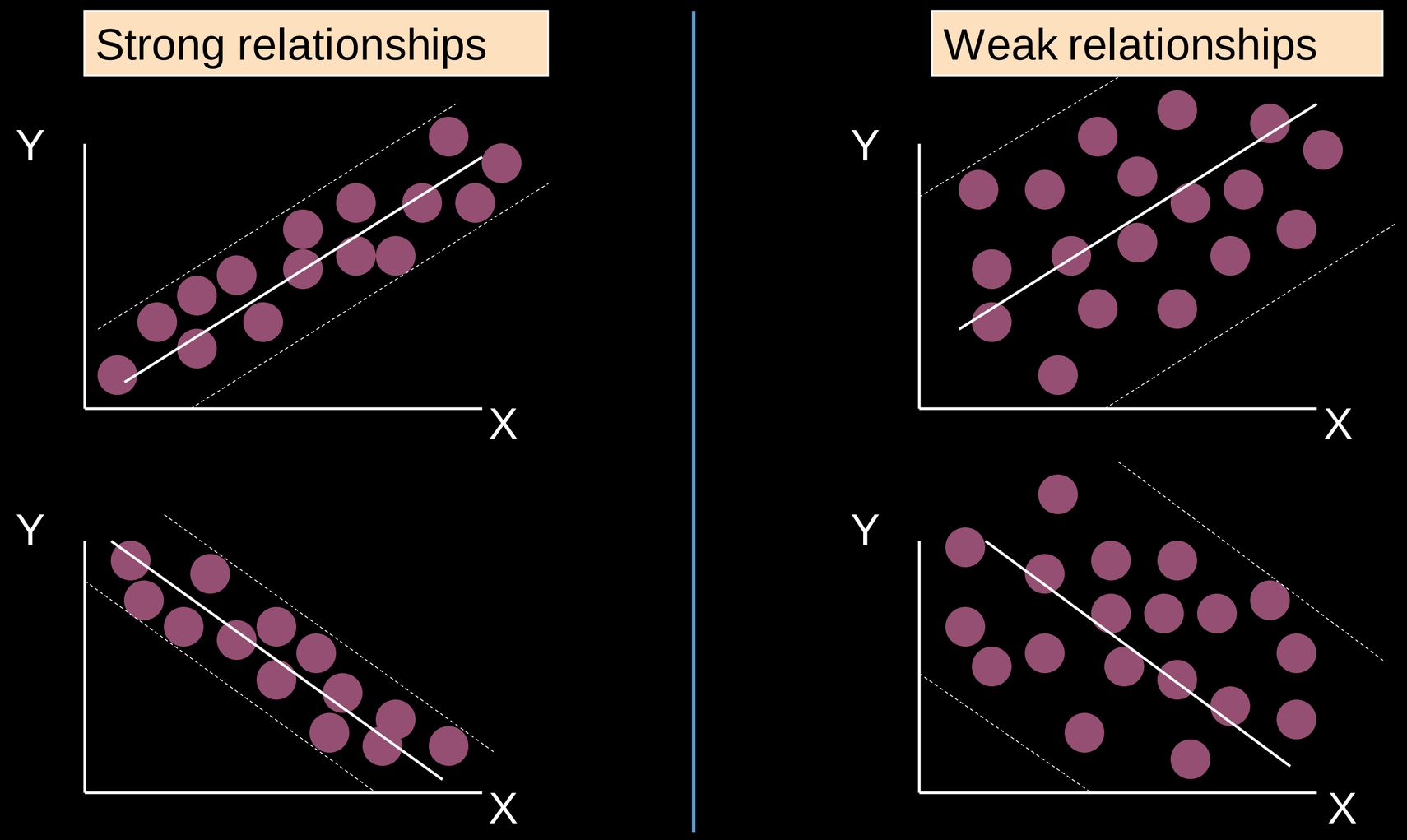
Curvilinear relationships





- Slide from: Statistics for Managers Using Microsoft® Excel 4th Edition, 2004 Prentice-Hall

Linear Correlation



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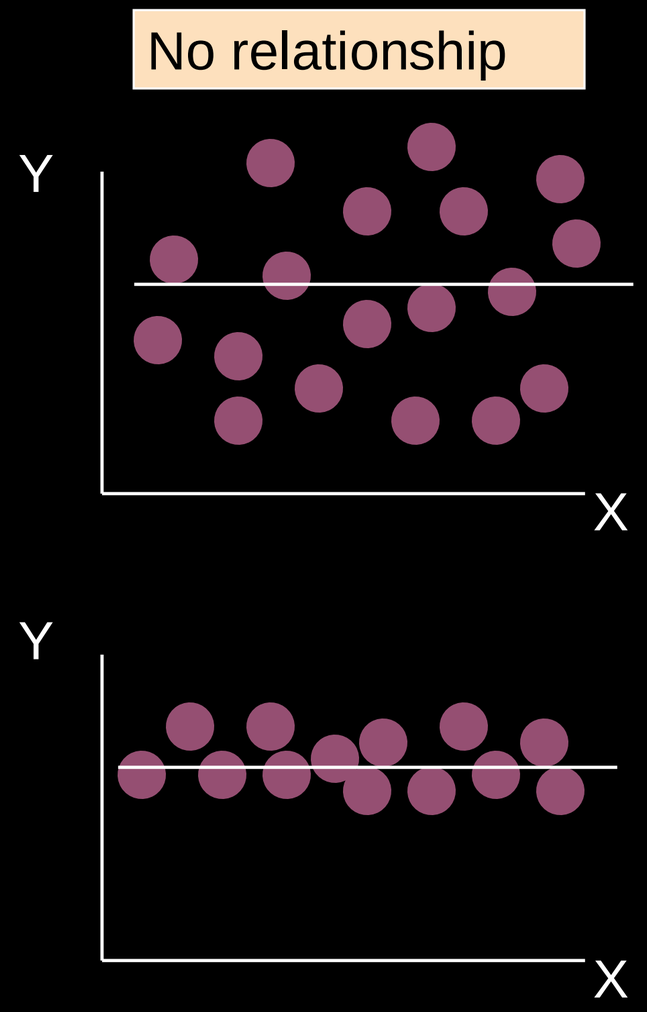
Linear Correlation



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THANK YOU